

CASE RECORDS of the MASSACHUSETTS GENERAL HOSPITAL

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Case 1-2026: A 50-Year-Old Woman with Fever and Abdominal Pain

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PRESENTATION OF CASE

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CME



Dr. Christina Shincovich (Medicine): A 50-year-old woman was admitted to this hospital because of fever and abdominal pain after she had returned from a trip to rural Brazil.

The patient was born and raised in Brazil and immigrated to Boston in early adulthood. In the weeks leading up to the current presentation, she traveled to a remote area in northeastern Brazil. Four days before the current presentation, after she had returned to Massachusetts, fatigue developed. The next day, she had several episodes of watery diarrhea that were accompanied by nausea and mild diffuse abdominal discomfort. Her symptoms lasted for 3 days, and her weight decreased by 3 kg. Dyspnea and lethargy developed, and she presented to the emergency department of this hospital.

In the emergency department, the patient answered questions only intermittently owing to malaise; additional history was obtained from her husband. In the days before the current presentation, the patient had noted chills, malaise, headache, and pain in the knees and legs. She had not noted rash, jaundice, pruritus, chest pain, cough, eye or ear pain or dysfunction, hematochezia, melena, dysuria, or hematuria.

The patient had a history of schistosomiasis that had occurred during childhood and had been complicated by noncirrhotic portal hypertension, esophageal varices, splenomegaly, chronic thrombocytopenia, and chronic leukopenia. Other medical history included gastroesophageal reflux disease, iron deficiency anemia, hypothyroidism, and allergic rhinitis.

Approximately 10 years before the current presentation, ultrasonography of the liver reportedly revealed coarse echogenicity that was consistent with fatty infiltration. Evaluation for a cause of liver disease was unrevealing, and a liver biopsy showed features of nodular regenerative hyperplasia but no clinically significant fibrosis. Six years before the current presentation, liver elastography showed a liver stiffness of 18.4 kPa (reference range, 2 to 7), a finding consistent with F4

fibrosis (liver stiffness, >14 kPa). The levels of thyrotropin and alpha-fetoprotein were normal; other laboratory test results obtained at that time are shown in Table 1. Four years before the current presentation, liver ultrasonography reportedly revealed morphologic features in the liver that were consistent with cirrhosis, as well as gallbladder wall thickening and splenomegaly, with the spleen measuring 16 cm in the greatest dimension (reference value, ≤12). Esophagogastroduodenoscopy procedures performed 30 months and 2 months before the current presentation reportedly showed grade II esophageal varices (2 to 3 mm in diameter and mildly tortuous or straight).

Medications included levothyroxine, cetirizine, ergocalciferol, resveratrol, and a multivitamin. Propranolol had caused fatigue, and there was a history of adverse reactions to medications containing sulfa. The patient had received vaccinations against severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2), hepatitis A and B viruses, and pneumococcus.

The patient lived in a suburb of Boston with her family, and she worked in the service industry, where she was often exposed to dust. She did not use alcohol, tobacco, or other substances. She had no history of travel other than the recent trip to Brazil. During the trip, she had been bitten by multiple mosquitoes while spending time at a beach in a region where cases of dengue fever and chikungunya virus infection were known to have occurred. She had not taken malaria prophylaxis. She had no known sick contacts. Information on other environmental, animal, and food exposures was not available. One of her brothers had died in his fourth decade of life from chronic liver disease attributed to alcohol use. Two other brothers had a history of hematemesis, one of whom had had schistosomiasis and had undergone splenectomy.

On examination, the temporal temperature was 38.4°C, the heart rate 105 beats per minute, the blood pressure 76/44 mm Hg, the respiratory rate 32 breaths per minute, and the oxygen saturation 100% while the patient was breathing ambient air. She appeared lethargic and unwell, with diaphoretic pale skin and labored rapid breathing. Petechiae of the posterior palate were present. The abdomen was slightly distended and diffusely tender; the tenderness was most notable

in the right upper quadrant and suprapubic area. Murphy's sign was absent, and the spleen tip was palpable. No costovertebral tenderness was noted. Trace edema of the legs was present, but there was no rash. The remainder of the examination was normal.

The high-sensitivity troponin T level was undetectable, and the creatine kinase level was normal; other laboratory test results are shown in Table 1. The urinary level of human chorionic gonadotropin was undetectable. A respiratory viral panel performed on a nasopharyngeal swab sample was negative for SARS-CoV-2, adenovirus, influenza viruses A and B, parainfluenza viruses 1 through 4, respiratory syncytial virus, rhinovirus, and enterovirus. Tests for bacteria such as *Bordetella pertussis*, *Chlamydia pneumoniae*, and *Mycoplasma pneumoniae* were also negative. Blood cultures were obtained.

During the patient's first hour in the emergency department, 1 liter of lactated Ringer's solution and vancomycin, ceftriaxone, and azithromycin were administered intravenously, and the blood pressure rose to 90/54 mm Hg. Electrocardiography showed sinus tachycardia and borderline-low QRS voltage in the precordial leads. A rapid malaria antigen test was negative.

Dr. Mark A. Anderson: Chest radiography (Fig. 1A) revealed diffuse interstitial and airspace opacities in both lungs.

Dr. Shincovich: Piperacillin–tazobactam was administered intravenously. During the next 2 hours, despite the administration of an additional 2 liters of intravenous crystalloid fluid, the blood pressure decreased to 73/41 mm Hg; treatment with a continuous infusion of norepinephrine was started at a rate of 6 µg per minute. During the next hour, oxygen was administered through a nasal cannula at a rate of 2 liters per minute, the norepinephrine dose was increased to 15 µg per minute, and treatment with a continuous infusion of vasopressin was added.

Dr. Anderson: Computed tomography (CT) of the chest (Fig. 1B and 1C), performed after the administration of intravenous contrast material in accordance with a pulmonary-embolism protocol, showed no evidence of pulmonary embolism. Findings included diffuse septal thickening, bilateral pleural effusions, diffuse edema in the mediastinum, a small pericardial effusion, and subcentimeter mediastinal and hilar lymph nodes.

Table 1. Laboratory Data.*

Variable	Reference Range, Other Hospital	6 Yr before Current Presentation, Other Hospital	Reference Range, Adults, This Hospital†	On Current Presentation, This Hospital	On Arrival in ICU, This Hospital
Blood					
Hemoglobin (g/dl)	11.5–16.4	13.0	12.0–16.0	12.3	12.1
Hematocrit (%)	36.0–48.0	39.2	36.0–46.0	37.2	37.8
White-cell count (per μ l)	4000–10,900	2160	4500–11,000	2150	22,850
Differential count (per μ l)					
Neutrophils	1920–7600	1430	1800–7700	1750	19,130
Lymphocytes	720–4100	520	1000–4800	230	550
Monocytes	160–1100	140	200–1200	20	750
Eosinophils	0–500	60	0–900	0	180
Bands (%)	—	—	0–10	13.4	14.6
Peripheral-blood smear	—	—	No abnormal forms present	Plasma cells, metamyelo- cytes, and polychromatic red cells present	Metamyelocytes, vacuolated white cells, burr cells, and poly- chromatic red cells present
Platelet count (per μ l)	150,000–450,000	56,000	150,000–400,000	37,000	29,000
Sodium (mmol/liter)	136–145	141	135–145	138	137
Potassium (mmol/liter)	3.4–5.0	3.8	3.4–5.0	4.0	3.8
Chloride (mmol/liter)	98–107	105	98–108	104	102
Carbon dioxide (mmol/liter)	22–31	26	23–32	20	10
Urea nitrogen (mg/dl)	6–23	16	8–25	21	24
Creatinine (mg/dl)	0.50–1.20	0.49	0.60–1.50	0.97	1.33
Glucose (mg/dl)	70–115	125	70–110	91	21
Calcium (mg/dl)	8.6–10.7	8.9	8.5–10.5	8.4	7.7
Magnesium (mg/dl)	—	—	1.7–2.4	1.6	1.8
Lactate (mmol/liter)	—	—	0.5–2.0	4.7	13.2
Total protein (g/dl)	6.0–8.0	7.1	6.0–8.3	6.1	5.1
Albumin (g/dl)	3.5–5.2	3.8	3.3–5.0	3.6	3.1
Alanine aminotransferase (U/liter)	10–50	125	7–33	76	102
Aspartate aminotransferase (U/liter)	10–50	101	9–32	64	131
Alkaline phosphatase (U/liter)	40–130	229	45–115	242	396
Total bilirubin (mg/dl)	0.0–1.0	0.9	0.0–1.0	2.2	5.7
Direct bilirubin (mg/dl)	—	—	0.0–0.4	1.2	4.4
Fibrinogen (mg/dl)	—	—	200–400	159	79
Prothrombin time (sec)	11.8–15.2	14.8	10.0–13.0	20.9	27.8
International normalized ratio	0.9–1.1	1.2	0.9–1.1	1.9	2.5
Partial-thromboplastin time (sec)	—	—	24.0–37.5	54.1	79.7
D-dimer (ng/ml)	—	—	<500	—	>7650
Haptoglobin (mg/dl)	—	—	30–200	—	20
Lactate dehydrogenase (U/liter)	—	—	110–210	—	277
Venous pH	—	—	7.30–7.40	—	7.15

Table 1. (Continued.)

Variable	Reference Range, Other Hospital	6 Yr before Current Presentation, Other Hospital	Reference Range, Adults, This Hospital [†]	On Current Presentation, This Hospital	On Arrival in ICU, This Hospital
Urine					
Color	—	—	Yellow	—	Orange
Clarity	—	—	Clear	—	Turbid
pH	—	—	5.0–9.0	—	6.0
Specific gravity	—	—	1.001–1.035	—	1.020
Glucose	—	—	Negative	—	Negative
Ketones	—	—	Negative	—	Trace
Leukocyte esterase	—	—	Negative	—	1+
Nitrite	—	—	Negative	—	Negative
Blood	—	—	Negative	—	3+
Protein	—	—	Negative	—	2+
Erythrocytes (per high-power field)	—	—	0–2	—	20–50
Leukocytes (per high-power field)	—	—	<10	—	>100
Bacteria	—	—	None	—	1+

* To convert the values for urea nitrogen to millimoles per liter, multiply by 0.357. To convert the values for creatinine to micromoles per liter, multiply by 88.4. To convert the values for glucose to millimoles per liter, multiply by 0.05551. To convert the values for calcium to millimoles per liter, multiply by 0.250. To convert the values for magnesium to millimoles per liter, multiply by 0.4114. To convert the values for lactate to milligrams per deciliter, divide by 0.1110. To convert the values for bilirubin to micromoles per liter, multiply by 17.1. ICU denotes intensive care unit.

[†] Reference values are affected by many variables, including the patient population and the laboratory methods used. The ranges used at Massachusetts General Hospital are for adults who are not pregnant and do not have medical conditions that could affect the results. They may therefore not be appropriate for all patients.

CT of the abdomen and pelvis (Fig. 1D and 1E) showed morphologic features in the liver that were consistent with cirrhosis. Large portosystemic collaterals were present, including gastroesophageal varices and a large patent periumbilical vein draining into the right common femoral vein. The spleen measured 18.8 cm in the greatest dimension. There was hydroureteronephrosis on the right side resulting from external compression at the level of the pelvic brim by an exophytic uterine mass, measuring 12.2 cm in the greatest dimension, that was consistent with a fibroid. Distal gastric, duodenal, and diffuse colorectal wall thickening was noted, along with ascites and diffuse mesenteric and retroperitoneal edema.

Dr. Shincovich: The patient was admitted to the medical intensive care unit (ICU) of this hospital 6 hours after presentation. Laboratory test results obtained at that time are shown in Table 1. A radial arterial catheter and a right internal jugular central venous catheter were placed, and the trachea was intubated for initiation of mechanical ventilation.

Dr. David M. Dudzinski: Electrocardiography showed sinus rhythm at a lower heart rate than previously noted, despite the worsening shock syndrome, as well as borderline-low QRS voltage in the precordial leads, subtle PR-segment depression in the inferior and apical leads, and J-point elevation with concave submillimeter ST-segment elevation in the inferior, apical, and lateral leads. Bedside cardiac ultrasonography showed globally reduced biventricular function, moderate tricuspid regurgitation, a plethoric inferior vena cava, and no pericardial effusion.

Dr. Shincovich: Vancomycin, cefepime, metronidazole, dextrose, and propofol were administered intravenously.

A diagnostic test was performed, and management decisions were made.

DIFFERENTIAL DIAGNOSIS

Dr. Stephen C. Threlkeld: This 50-year-old woman with a history of portal hypertension and liver disease had fatigue, diarrhea, leg pain, headache, and chills after returning from a trip to remote

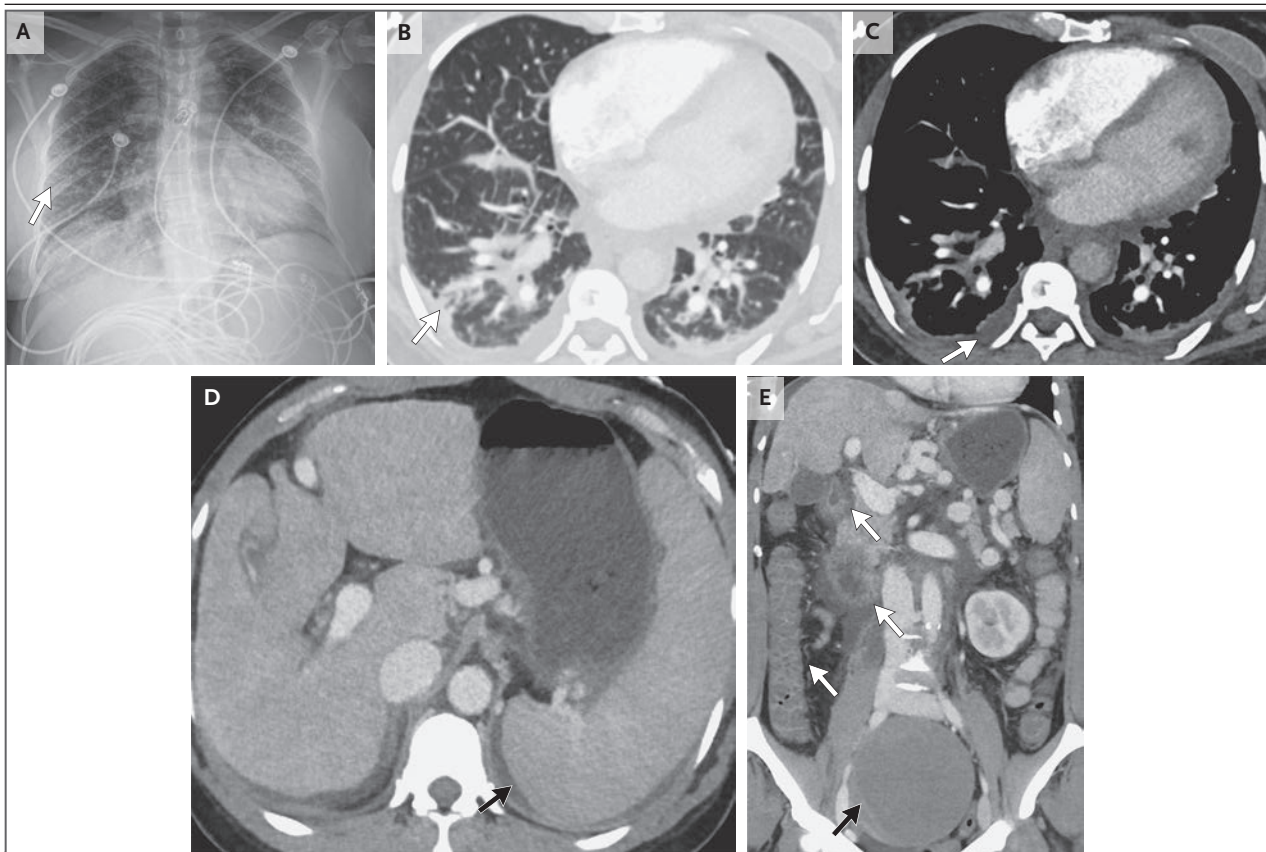


Figure 1. Radiograph and CT Images.

An upright chest radiograph (Panel A) and axial images from contrast-enhanced CT of the chest (Panels B and C) show diffuse interstitial and airspace opacities in both lungs (Panels A and B, arrows) and small pleural effusions (Panel C, arrow). An axial image from non-contrast-enhanced CT of the abdomen and pelvis (Panel D) and a coronal reconstruction image (Panel E) show morphologic features in the liver that are consistent with cirrhosis; large portosystemic collaterals, including gastroesophageal varices; splenomegaly (Panel D, arrow); a 12.2-cm exophytic uterine mass (Panel E, black arrow); distal gastric, duodenal, and diffuse colorectal wall thickening (Panel E, white arrows); and small-volume ascites and diffuse mesenteric and retroperitoneal edema.

northeastern Brazil. Refractory shock ensued, with disseminated intravascular coagulation (DIC), capillary leak, and marked leukocytosis with a left shift. In working toward the most likely diagnosis in this patient's case, I will focus on the distinctive feature of rapid deterioration with refractory septic shock and DIC, while considering her preexisting diseases and recent travel.

HOST FACTORS

The patient had acquired schistosomiasis as a child in Brazil. The infection had been complicated by portal hypertension and hypersplenism, which can cause chronic leukopenia and thrombocytopenia. Although schistosomiasis can cause presinusoidal liver disease, the patient later had imaging findings that were consistent with fatty liver disease, fibrosis, and possible cirrhosis; these

features were not confirmed on biopsy. Cirrhosis leads to several important functional immunologic defects, including hypocomplementemia, increased gut translocation of bacteria, and portosystemic shunting.^{1,2}

INFECTIONS RELATED TO RECENT TRAVEL

The patient's recent travel to Brazil increased her risk of exposure to pathogens that can cause fulminant infection.

Bacterial Infections

Enteric fever due to *Salmonella enterica* serovar Typhi infection is the underlying cause in many cases of severe illness that occur after travel to Brazil, but the rapid progression of disease that was seen in this patient would be unusual with this diagnosis. In addition, the patient had no evi-

dence of relative bradycardia, no skin findings that were consistent with rose spots, and no imaging findings such as intestinal perforation that would be suggestive of the disease.

Nontyphoidal salmonella infection is also a consideration in this patient, in part because of its known association with schistosomiasis.^{3,4} If schistosoma reinfection had occurred while the patient was in Brazil, the resulting illness would most likely be mild because of her partial immunity, but associated salmonella bacteremia could cause a more fulminant presentation.

Vibrio vulnificus infection is often characterized by a fulminant presentation in patients with advanced liver disease after exposure to seawater or undercooked seafood. Infection with other vibrio species can also manifest with acute sepsis.⁵ However, this patient did not have the hemorrhagic bullous skin lesions that are characteristic of vibrio infection, and infection with any vibrio species has a very short incubation period, which makes this diagnosis unlikely.

Listeria infection can cause gastroenteritis that is complicated by sepsis in immunocompromised patients, particularly those who have consumed unpasteurized dairy products.

Parasitic Infections

Malaria — mainly malaria due to *Plasmodium falciparum* infection — is always a consideration in travelers with fever and severe illness who are returning from regions where the disease is endemic, particularly those who did not take prophylaxis. However, malaria is very uncommon in northeastern Brazil, which is more arid than other regions. In addition, more than 90% of malaria cases are attributed to *P. vivax* infection, which is less likely to cause fulminant deterioration than *P. falciparum* infection.⁶ It is notable that a rapid malaria antigen test was negative in this patient.

Acute Chagas' disease, which is due to *Trypanosoma cruzi* infection, can occur in Brazil after consumption of sugarcane juice contaminated with triatomine bugs. However, disease progression is typically much slower than that seen in this patient, and the disease is often associated with myocarditis or meningoencephalitis.⁷

Viral Infections

Arbovirus infections are common in Brazil. Infection with Zika virus or with chikungunya virus seldom progresses to septic shock. Oropouche

virus disease, an emerging infection in Brazil, is usually a much milder illness than that seen in this patient, although fulminant cases have been reported.⁸

Dengue virus infection is an essential consideration in this patient, particularly given its notable increase in prevalence in recent years. Dengue shock syndrome typically occurs in patients who are reinfecting with a dengue virus of a different serotype and have antibody-mediated enhancement of disease, with deterioration over a period of hours to DIC and multiple-organ failure.⁹ However, rapid progression from leukopenia to marked leukocytosis with a left shift makes any viral cause unlikely in this patient.

INFECTIONS UNRELATED TO RECENT TRAVEL

Although it is tempting to speculate that this patient had acquired a tropical disease during her recent trip, we should also consider diseases that are unrelated to her travel. Babesia infection is common in New England, can have a month-long or longer incubation period, and can have severe manifestations in patients with splenic dysfunction.

Pyelonephritis may be associated with septic shock, and this patient's imaging studies showed evidence of chronic ureteral obstruction. However, pyelonephritis is unlikely in this case, given that there were no localizing symptoms or inflammatory findings on CT.

Neisseria meningitidis infection is a classic cause of fulminant sepsis and DIC. The typical petechial or hemorrhagic rash is not present in all cases. Leg pain can be a presenting symptom in patients with meningococemia. Invasive meningococcal disease caused by serogroup W has been characterized by a lower likelihood of purpuric rash and a higher incidence of abdominal pain and diarrhea than disease caused by other serogroups, differences that may lead to delays in diagnosis.¹⁰⁻¹² Splenic dysfunction and possible cirrhosis, which leads to hypocomplementemia, may be risk factors for meningococemia in this patient.

Finally, enteric gram-negative bacteria are the most common cause of infection in patients with advanced chronic liver disease. In affected patients, associated immune dysfunction, increased gut translocation, and portosystemic shunting may increase the likelihood of septic shock.¹

Septic shock caused by infection with gram-negative bacteria is the most likely diagnosis in

this patient. I suspect that the blood cultures grew enteric gram-negative rods or *N. meningitidis*.

DR. STEPHEN C. THRELKELD'S
DIAGNOSIS

Rapidly progressive shock in a returning traveler, most likely septic shock caused by infection with a gram-negative pathogen.

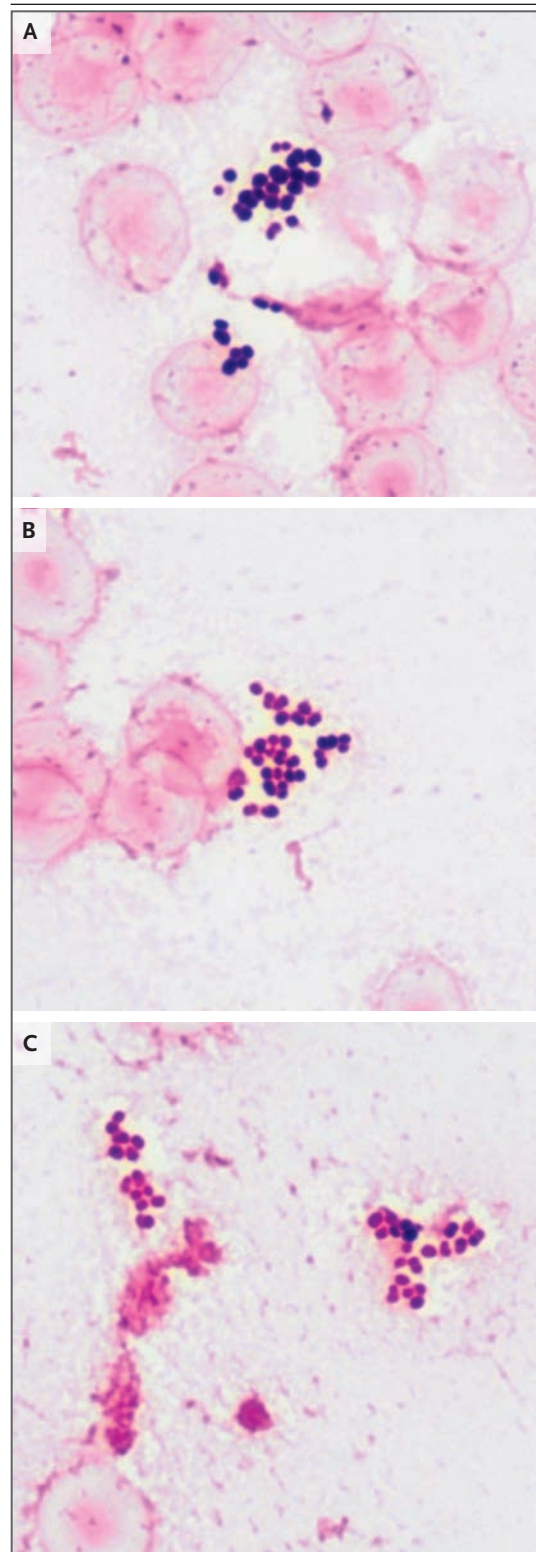
DIAGNOSTIC TESTING

Dr. Melis N. Anahtar: The diagnostic test in this case was a blood culture. Bacterial growth was detected in the aerobic bottle after 21 hours of incubation. Gram's staining was performed on the positive blood-culture broth, and the bacteria initially appeared to be gram-positive cocci in clusters (Fig. 2A). The broth was subcultured on agar plates for organism identification and susceptibility testing. Examination of the blood agar plate after 10 hours of incubation revealed small, gray, glistening colonies. Organism identification performed with matrix-assisted laser desorption ionization–time of flight (MALDI-TOF) mass spectrometry showed findings consistent with *N. meningitidis*.

This preliminary identification prompted re-examination of results from the initial Gram's staining, as well as repeat Gram's staining performed on broth from the same bottle with progressively increased decolorization time (Fig. 2B and 2C). With full decolorization, the bacteria appeared to be gram-negative diplococci, and the report was corrected with identification of the organism as *N. meningitidis*. Antimicrobial susceptibility testing performed with the gradient diffusion method (Etest) showed susceptibility to penicillin G, ceftriaxone, and ciprofloxacin.

Figure 2. Blood-Culture Specimens.

On initial Gram's staining of broth from the positive blood-culture bottle (Panel A), the bacteria appear to be gram-positive cocci in clusters. On repeat Gram's staining of broth from the same blood-culture bottle with progressively increased decolorization time (Panels B and C), removal of the crystal violet–iodine complex is shown, and the bacteria appear to be gram-negative diplococci. Images are courtesy of Emily Williams.



The Kirby–Bauer disk-diffusion method showed susceptibility to rifampin. The isolate was sent to the state public health laboratory for serotyping, which revealed serogroup W135.

Gram's staining is a robust and fundamental method of microbiologic diagnostic testing, but it is not infallible. Characterization of a bacterial species on Gram's staining is often thought of as binary — either gram-positive or gram-negative, depending on whether the crystal violet–iodine complex is retained or removed, respectively, from the cell with alcohol treatment, or “decolorization.” However, characterization of an organism on Gram's staining depends on the staining technique and the thickness and composition of the cell wall and cell membrane, which can vary according to the bacterial growth phase and environmental stressors.^{13,14} Some of these complexities can be avoided by implementing rapid molecular diagnostic testing that provides genus- and species-level identification of bacteria directly from positive blood cultures in just over an hour.¹⁵ However, it is notable that *N. meningitidis* infection is so uncommon in the United States that the pathogen is not included on all commercially available rapid molecular panels.

MICROBIOLOGIC DIAGNOSIS

Neisseria meningitidis bacteremia.

DISCUSSION OF MANAGEMENT

Dr. George A. Alba: This patient presented with septic shock caused by life-threatening organ dysfunction due to a dysregulated host response to infection.¹⁶ After crystalloid resuscitation, the patient had evidence of capillary leak, pulmonary edema, and myocardial dysfunction. The administration of first-line vasoactive agents norepinephrine and vasopressin was prioritized to maintain organ perfusion and prevent iatrogenic fluid overload, which has been associated with worse outcomes in patients with sepsis.^{17,18} The patient was treated with broad-spectrum antimicrobial therapy, and cross-sectional imaging did not identify a source of sepsis that was amenable to intervention.^{19,20}

Unfortunately, refractory shock developed.²¹

We administered epinephrine as a third vasoactive agent, given the patient's underlying myocardial dysfunction; a bicarbonate infusion to buffer her worsening metabolic acidosis, which impairs the efficacy of catecholamine vasopressors and further depresses myocardial function²²; and stress-dose glucocorticoid therapy with hydrocortisone and fludrocortisone.²³ The doses of vasoactive agents were escalated despite these interventions, and we administered hydroxocobalamin, which acts as a nitric oxide scavenger and results in potent vasoconstriction.²⁴ Venoarterial extracorporeal membrane oxygenation can be considered in select cases of refractory septic shock, but this rescue strategy was not pursued in this patient owing to the presence of profound coagulopathy.

Dr. Nesli Basgoz: Invasive meningococcal disease most frequently manifests as one of three syndromes: meningitis alone, meningitis with meningococcemia, or acute meningococcemia alone. Because clinical manifestations vary, a high index of suspicion is required.

Risk factors for invasive meningococcal disease include hypocomplementemia (acquired or iatrogenic, due to certain medications), human immunodeficiency virus infection, functional asplenia, travel exposure, and occupational exposure. This patient had a history of chronic liver disease and possible cirrhosis. Because complement is made in the liver, patients with cirrhosis may have hypocomplementemia. Portal hypertension, hypersplenism, and malnutrition are also associated with an increased risk of bacteremia and poor clearance.

The treatment of choice for invasive meningococcal disease is the β -lactam antibiotic agent ceftriaxone. Resistance to this agent is uncommon, but given the critical nature of the disease, antibiotic susceptibility testing is performed.

Dr. Angela G. Fowler: Invasive meningococcal disease is associated with a substantial public health burden worldwide, with an estimated 1.2 million cases and 135,000 deaths occurring each year. Mortality remains high despite treatment with antibiotics. The case fatality rate in the United States is estimated to be 5 to 15%.^{25,26}

In the United States, the incidence of invasive meningococcal disease is highest among infants and young children. However, a growing propor-

tion of cases are occurring in older age groups. Serogroup B is the most common serotype in young children, and serogroups C, Y, and W are more prevalent in older persons, such as this patient. The incidence of invasive meningococcal disease decreased from 2006 to 2021, but it has increased sharply since 2021, primarily owing to serogroup Y disease.²⁷

Before becoming ill, this patient had visited Brazil, where invasive meningococcal disease is also a nationally notifiable disease, and serotype data are reported when available. In Brazil, the annual incidence of invasive meningococcal disease from 2010 through 2021 was 0.59 cases per 100,000 persons, with the incidence generally decreasing over that period.²⁸ There was also a marked change in the serogroup distribution. Serogroup C was the predominant serotype until the introduction of the meningococcal C vaccination program for infants in 2010 and the introduction of the meningococcal C booster dose for adolescents in 2017. Since the initiation of these vaccination programs, the incidence of serogroup C disease has gradually decreased, and the incidence of serogroup B and W infections has increased.

One of the most important public health measures for the reduction of invasive meningococcal disease is vaccination. Meningococcal ACWY vaccines are composed of capsular polysaccharides conjugated to a protein carrier. Before the changes this week, the Centers for Disease Control and Prevention recommended that children receive a meningococcal ACWY vaccine at 11 or 12 years of age and a booster dose at 16 years of age.²⁹ In addition to routine vaccination of adolescents, meningococcal vaccination is indicated for younger children and adults with an increased risk of meningococcal disease due to medical conditions, occupation, or travel.

Another key component of the public health response to invasive meningococcal disease is the use of antibiotic prophylaxis for the prevention of secondary attacks in persons who have been in contact with an infected patient. Close contacts of a patient with meningococcal disease — household members, childcare providers, and others who have been directly exposed to the infected patient's oral secretions — should receive antibiotic prophylaxis as soon as possible after exposure, regardless of vaccination status. Health care

workers should receive prophylaxis if they are involved in airway management in a patient with meningococcal disease or have been exposed to the patient's respiratory secretions. Options for antibiotic prophylaxis are rifampin, ciprofloxacin, ceftriaxone, and azithromycin.

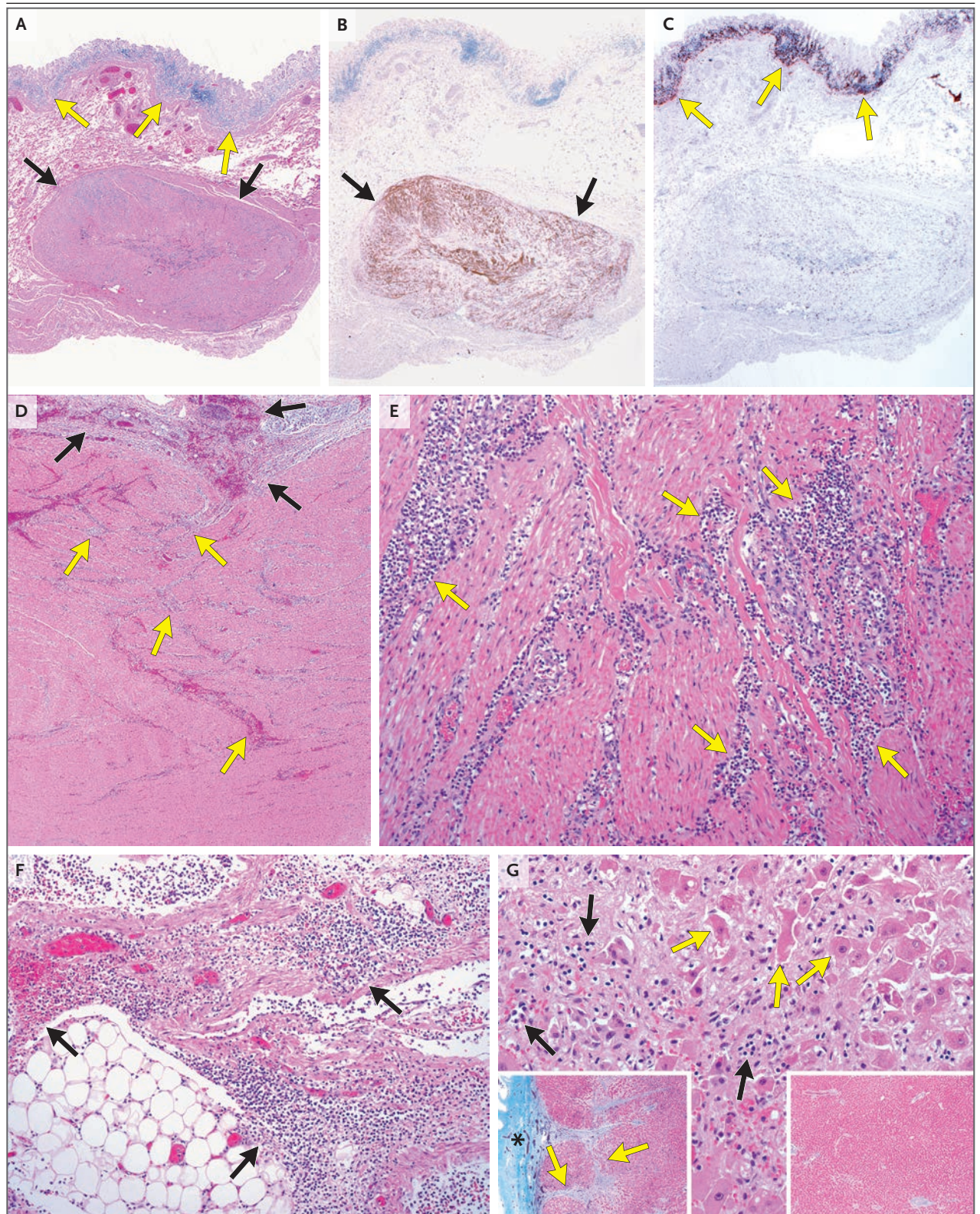
Dr. Shincovich: After the development of refractory shock in this patient, goals of care were discussed with her family, and the decision was made not to pursue resuscitation in the event of cardiac arrest. Six hours after the patient's admission to the ICU, and 12 hours after presentation to the emergency department, a pulseless electrical activity event occurred, and the patient died. The previously described diagnostic blood culture was found to be positive several hours after her death, confirming the diagnosis of disseminated meningococcal disease.

AUTOPSY

Dr. James R. Stone: An autopsy was performed. Small fibrin-rich thrombi were present in the lungs, kidneys, colon, and heart, findings consistent with DIC. The gastric body contained two small gastrointestinal stromal tumors and evi-

Figure 3 (facing page). Autopsy Specimens.

Hematoxylin and eosin staining of a section of the body of the stomach (Panel A) shows a mass in the submucosa (black arrows) and chronic inflammation in the mucosa (yellow arrows). Immunohistochemical staining for CD117 (Panel B) is positive in tumor cells (in brown; arrows), a finding that indicates a gastrointestinal stromal tumor. Immunohistochemical staining for CD3 (Panel C) shows numerous T lymphocytes in the mucosa (in brown; arrows). Hematoxylin and eosin staining of a section of the pylorus of the stomach (Panels D and E) shows hemorrhage and inflammation in the submucosa (black arrows) and muscularis propria (yellow arrows). The mucosa was intact with focal acute inflammation (not shown). Hematoxylin and eosin staining of a section of perigastric adipose tissue (Panel F) shows extension of the acute inflammation (arrows). Hematoxylin and eosin staining of a section of the subcapsular portion of the liver (Panel G) shows hepatocyte loss (in the center) with remaining hepatocytes (yellow arrows) and chronic inflammation (black arrows). Trichrome staining of the liver (Panel G, insets) shows capsular fibrosis (left inset, asterisk) and focal bridging fibrosis (left inset, arrows) in the subcapsular regions but no substantial fibrosis in the rest of the liver (right inset).



dence of chronic gastritis (Fig. 3A, 3B, and 3C); special staining was negative for *Helicobacter pylori*. The gastric pylorus had wall thickening (up to 0.8 cm) on gross examination. Histologic examination revealed acute transmural phlegmonous gastritis with acute inflammation and hemorrhage involving the submucosa and muscularis propria (Fig. 3D and 3E). Acute inflammation extended into the perigastric soft tissue (Fig. 3F) and was associated with local lymphadenitis. Scarring in the pancreas was consistent with chronic pancreatitis, as well as focal mild acute pancreatitis. Special staining was negative for microorganisms. Sections of the colon showed no evidence of colitis.

The liver was small and had a centrilobular hepatocellular injury pattern with hepatocyte dropout (Fig. 3G). Trichrome staining revealed capsular fibrosis and focal bridging fibrosis that was limited to the subcapsular regions, with relative sparing of the internal parenchyma. Features of active schistosomiasis and overt cirrhosis were not identified. The spleen was enlarged, with changes consistent with portal hypertension. The uterus contained a leiomyoma, measuring 13.5 cm in the greatest dimension, that was arising from the right wall and was associated with compression and proximal dilatation of the right ureter. The lungs were edematous without evidence of pneumonia, and the brain had no pathologically significant abnormalities.

The pathological findings were indicative of phlegmonous gastritis with extension into the perigastric soft tissue that had resulted from an infectious process. The pathological changes associated with phlegmonous gastritis have been described at autopsy or after surgical gastrectomy in a small number of cases.³⁰⁻³⁵ There is usually focal or diffuse thickening of the gastric wall on

gross examination, which is associated with hemorrhage, edema, and neutrophilic abscesses on histologic examination. Extensive necrosis, leading to focal wall thinning, may be present. Discoloration, erosion, or ulceration of the mucosa may be noted. Although phlegmonous gastritis is most commonly associated with streptococcal infection, it has also been reported with gram-negative bacterial infections.³⁶ Serogroup W meningococcal infection has been associated with death occurring after an acute gastrointestinal presentation, although the pathological changes have not been well understood.³⁷

PATHOLOGICAL DIAGNOSIS

Meningococcal infection complicated by phlegmonous gastritis and disseminated intravascular coagulation.

FINAL DIAGNOSIS

Disseminated meningococcal disease.

This case was presented at the Harvard Medical School postgraduate course "Infectious Diseases in Adults 2025," directed by Nesli Basgoz, Ruanne V. Barnabas, Rajesh T. Ghandi, and Sandra B. Nelson.

Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

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